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COVER STORY

Why it's important to let sediment reach the shore

Photo of the Lower Cook Inlet of Alaska's Kachemak Bay by National Oceanic and Atmospheric Administration (NOAA), public domain

Sediment comes in different sizes.

Lance Frazer · Ensia

In the Benediction for his 1968 autobiographical book, *Desert Solitaire: A Season in the Wilderness*, American author, essayist and environmental advocate Edward Abbey wrote, “May your rivers flow without end, meandering through pastoral valleys.” Today, many of those pastoral valleys have become homes to dams — altering the flow of rivers, sometimes ending it prematurely, and creating issues downstream we have only begun to understand.

In June 2022, Evan Dethier, then a postdoctoral fellow at Dartmouth, led a study using satellite images from the NASA/USGS Landsat Program and digital hydrologic data records from 1984 to 2020 that found that dam building in the 20th century had caused the rivers of North America, Europe/Eurasia and Asia to deliver 49% less suspended sediment to the oceans.

“Humans have been able to alter the world’s biggest rivers at rates that are unprecedented in the recent geological record,” Dethier said in a release of the study’s findings.

In 2019, hydropower, including that generated by dams, accounted for an estimated 60% of all renewable electricity. Dams are important for preventing floods, and they provide water for domestic and industrial use; it has been estimated that as much as 40% of irrigated land globally relies on dams. The Hoover Dam generates electricity for 1.3 million people in California, Nevada and Arizona; provides water for 16 million people; and irrigates more than 1.5 million acres (600,000 hectares) of land.

But, says Dethier, those benefits come with downsides.

“Dams alter the seasonal passage of water,” he says. “They also block the passage of fish and other important organisms. And, as we continue to learn, they trap vast amounts of sediment.”

To understand why that’s important, we need to take a closer look at sediment.

Biodiversity Bed

A 2011 estimate put the amount of sediment trapped behind the world’s dams at 73 cubic kilometers (18 cubic miles) — more than two times the maximum storage capacity of Lake Mead, the reservoir formed by the Hoover Dam and the largest reservoir by volume in the U. S. It’s estimated that by the middle of this century, dams worldwide will lose about 26% of their water storage capacity due to sediment buildup.

The flow of the river and the sediment it carries creates the form of the river and how it will migrate over hundreds to thousands of years,” says Jim Best, professor of sedimentary geology at the University of Illinois Urbana-Champaign. Photo of the Lower Cook Inlet of Alaska’s Kachemak Bay by National Oceanic and Atmospheric Administration (NOAA), public domain

Sediment comes in different sizes. Coarse sediment forms the “architecture” of most riverbeds, while fine-grained sediment is important for some floodplains and mudflats, as well as for conveying nutrients. “In general, we tend to trap coarser sediment and flush finer sediment downstream, and not replacing the coarser sediment can lead to the changing of the downstream environment, which can sometimes exacerbate the risk of flooding,” says Jim Best, professor of sedimentary geology at the University of Illinois Urbana-Champaign.

“The flow of the river and the sediment it carries creates the form of the river and how it will migrate over hundreds to thousands of years,” he says. “Rivers also provide ecological niches for a wide range of creatures, and the deposition of sediment and its sculpting creates those habitats. And the nutrients and organic materials it carries creates biodiversity in many of the world’s rivers.”

According to Brian Cluer, staff geomorphologist and research physical scientist for the West Coast Region of NOAA National Marine Fisheries Service, “many areas have been damaged or destroyed due to a lack of sediment. Now, one price we’re paying for that is endangered or valuable species damaged because of being denied habitat necessary for their life cycle. Salmon spawning beds are being damaged or destroyed, and processes that recharge floodplains are disrupted as some of the best farmlands are not being replenished with fresh sediment. Downstream, estuaries deprived of sediment are becoming more vulnerable to rising global sea levels.”

Dam Removal

Cluer was involved in the 2011 project to remove the Elwha Dam in Olympic National Park as part of the Elwha River Ecosystem and Fisheries Restoration Act, at the time the biggest dam removal in the U. S.

“The aftermath of the Elwha Dam removal shows the positive things that can happen when sediment is allowed to reach the mouth of the river. The beach area, once composed of volleyball-sized rocks, is now sandy and extends for hundreds of meters offshore. Clam fishing has returned, as well as salmon, and the extended shoreline provides protection from storms and sea level rise,” Cluer says. “When sediment is deposited, it creates the ability of the river to meander again, improving the richness and diversity of habitats.”

Meanwhile, Dethier says, the removal of dams along the Klamath River, which runs through Oregon and Northern California, “is an exciting test case for what happens to a river when four long-standing dams are removed. It will be interesting to follow the river’s recovery. Research tells us that if dams are removed properly, rivers can be incredibly resilient.”

Removing dams properly is key. While returning sediment downstream “is, in the long run, positive for the ecosystem, as it replenishes what was lost during the life of the dam and improves connectivity to floodplains, which we know is vital to river health,” Cluer says, he warns that this solution is “a large disturbance event, and there are impacts to existing habitats.” When assessing the impacts of dam removal, Cluer continues, much of the work involves predicting the short-term negative impacts and comparing them to predicted long-term benefits.

For example, Cluer says, “if the dam has stored sediments from an industrial area, the sediment could contain pollutants, and if it is contaminated, you get a polluted wave of sediment headed downstream, causing an impact on the river environment.”

One such example is highlighted in a study on the Mud River Reservoir in West Virginia, downstream from the mountaintop Hobet coal mine, which operated from 1974 to 2015. In the study, sediment cores showed increases in the storage of inorganic carbon, selenium and calcium, “which could have lasting impacts on reservoir biota and downstream environments long after reclamation efforts are applied.”

Other Solutions

In lieu of releasing sediment during dam removal, there are other ways to remove sediment buildup behind dams. For example, “dredging is most likely to be least impactful to downstream environments because flushing has much more negative consequences, such as increased turbidity, increased nutrient loading, higher heavy metal contamination and the like,” says Aubrey Hillman, an assistant professor in the Department of Atmospheric and Environmental Sciences at the University at Albany. However, getting the dredged sediment to areas where it’s needed can be expensive, especially if the dam is in a remote area, Cluer warns.

Other approaches include ways to bypass dams and feed water, fish and sediment back into rivers, such as a connecting channel around a reservoir as part of the Chimney Hollow Reservoir project in Colorado. Meanwhile, in Japan, a 4.3-kilometer (2.7-mile) bypass tunnel on the Miwa Dam completed in 2004 moves an average of 399,000 cubic meters (522,000 cubic yards) per year of mostly fine sediment, while coarse sediment is trapped upstream for use in construction.

The San Francisco Bay is home to hundreds of acres of baylands that support fish and wildlife, protect the shoreline and help buffer the impact of sea-level rise.

Lance Frazer

style="font-weight: 400;"> Rivers at Risk

style="font-weight: 400;">Two of the world's longest and most famous rivers, and the ecosystems and people that rely on them, are facing risks of river sediment not reaching deltas due to existing or proposed dams.

style="font-weight: 400;">According to Al Jazeera , the Nile River discharged more than 120 million metric tons (132 million tons) of sediment annually to its delta before the Aswan Dam was constructed in the 1960s. In recent years, sediment reaching the delta, which supports half of Egypt's economy and is home to more than 40% of its people, has dropped 98% , and estimates indicate that thousands of acres of the delta could be under water by century's end due to a combination of the missing sediment and rising sea levels. Water quality issues have increased as well due to the disruption of the river's natural flow, which historically flushed salts and contaminants from the delta.

style="font-weight: 400;">Additionally, the dam has helped lead to an increase in schistosomiasis, a disease caused by parasitic worms spread by freshwater snails that infects hundreds of millions of people worldwide and kills some 200,000 each year.

style="font-weight: 400;">"Society chose years ago to maximize short-term agriculture by disconnecting our rich floodplains from their river sediment sources," Cluer says. "Levees, dams, channelization, more people, more food, but no understanding of the massive looming problems due to the loss of floodplain-forming processes. Without reconnecting floodplain farmlands to the river and its sediment supply, the land and its soil are not long-term sustainable."

style="font-weight: 400;">Additionally, Cluer points to negative effects on aquifers around the world. "If you disconnect rivers from the floodplain, you deplete the underlying water supply. Disconnected floodplains then have to be irrigated, and when irrigation water comes from groundwater aquifers, we risk overdrafting them, and aquifer compaction results. And a compacted aquifer in the basin cannot be recovered," he says. "The compaction of global aquifer basins is irreversible."

class="wp-caption-text" id="caption-attachment-34959">In this pair of images showing the Three Gorges Dam on the Yangtze River in central China in partial completion in July 2000 (left) and in May 2006, sediment-filled water can be seen flowing through a channel in the earlier photo, but not by the time the dam spanned the entire river in 2006, when a large reservoir was formed behind it. Photo created by Jesse Allen, Earth Observatory, using ASTER data made available by NASA/GSFC/MITI/ERSDAC/JAROS, and U. S./ Japan ASTER Science Team

style="font-weight: 400;">On the other side of the world, the Amazon River Basin is roughly 7 million square kilometers (nearly 3 million square miles) — covering about 40% of

South America — and within its reaches, more than 300 dams are proposed that will impact the region and beyond.

style="font-weight: 400;">"These proposed projects threaten one of the world's greatest remaining resources, with biological, human and aesthetic consequences to this region," University of Illinois Urbana-Champaign's Best says. "Where we build a dam, we change the area's balance."

style="font-weight: 400;">In 2021, researchers including Suyog Chaudhari of the Michigan State University Department of Civil and Environmental Engineering published a study looking at the use of in-stream turbines in the Brazilian Amazon as an alternative to planned storage-based hydropower projects. The study estimated that "63% of the total energy planned to be generated from conventional hydropower in the Brazilian Amazon could be harnessed using in-stream turbines that use kinetic energy of water without requiring storage."

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style="font-weight: 400;">Given the large number of dams planned or proposed for the Amazon, what could be the impact on the second-longest river in the world?

style="font-weight: 400;">"When you disconnect the river from its floodplains, you impact carbon sequestration, you impact aquifer recharge, and you impact a vast natural ecosystem," says Cluer.

style="font-weight: 400;"> Not Just Rivers

style="font-weight: 400;">Other bodies of water are similarly affected. Hillman, whose focus has been on lakes across the country and overseas, says in her work she's found that "sedimentation rates have skyrocketed in the 20th century, and a big impact has been the number of dams that are on rivers feeding into lakes." When they do dredging or sluicing, she says, sedimentation rates shoot up fast.

style="font-weight: 400;">"Each lake is unique, and we walk a fine line between too much sediment and too little. Sediment is critical to keep a lake thriving, delivering things like nitrogen and phosphorus. We did a study in New York that found that more than 50% of phosphorus that comes into a lake comes through sediment." But, as the saying goes, too much of a good thing can be bad — in this case, too much sediment can help create a now-familiar scene of a lake covered in blue-green algae, or cyanobacteria. These algae blooms impact human recreation, kill aquatic species and more. "Much more research is needed on the interaction between rivers, lakes and dams," Hillman says.

style="font-weight: 400;">The San Francisco Bay is home to hundreds of acres of baylands that support fish and

wildlife, protect the shoreline and help buffer the impact of sea-level rise. Scott Dusterhoff, program managing director and senior scientist for the Resilient Landscapes Program at the San Francisco Estuary Institute, says the focus in an institute report called “Sediment for Survival” was to look at the impacts of lost sediment on marshes and tidal areas, a problem he describes as growing over time.

“In San Francisco Bay, the habitat provided by the marshes is super important,” says Jessica Lacy, research oceanographer with the USGS Pacific Coastal and Marine Science Center. “The marshes are also located in places where they’re protecting coastal communities.”

“We get a lot of sediment from the delta, which is fed by the San Joaquin and Sacramento rivers,” Dusterhoff says. He explains that sediment brought to the bay during the 1800s gold rush, through hydraulic mining, blowing up mountains and other approaches, moved the shoreline out into the bay, creating close to 200,000 acres (81,000 hectares) of tidal marsh. But, according to the report, “diking of baylands in the 19th and 20th centuries for agriculture, development, and commercial salt production resulted in an approximate 65% reduction in bayland extent compared to the early 19th century.” Dusterhoff goes on to explain that “more recently, the sediment supply to the bay from the delta has been decreasing, due, in large part, to all that gold mining working its way through the system and the building of large dams in the Sierra Nevada.”

With approximately 40% of the world’s population living within 100 kilometers (62 miles) of a coastline, what is happening to the San Francisco Bay area is an important example to pay attention to.

Hydraulic mining activity in the foothills of the Sierra Nevada, used in the 1800s to search for gold, brought sediment to the coast, moving the shoreline out into the bay. Photo public domain, USGS

According to Sediment for Survival, more than 450 million cubic yards (344 million cubic meters) of sediment will be needed by 2100 to maintain the bay’s current wetlands and mudflats, as well as areas planned for restoration. “The amount of sediment coming in now isn’t close to what’s needed to allow the shoreline to survive where it is,” Dusterhoff says. “It’s likely we won’t be able to save every tidal marsh, so there will be some tough choices to make.”

Learning to Do Better

In the face of such difficult decisions, humans should “take advantage of natural capital,” says Best. “Rivers provide floodplains for agriculture, environments for fish. They help provide energy, recreation and more, and we’ve been thriving in those areas for thousands

of years. Now we’re changing natural capital in ways that are just not sustainable.”

Cluer agrees. “Making ecosystems smaller and less resilient by disconnecting floodplains from rivers is a diminishing curve that citizens don’t realize is a big problem,” he says. “It takes a better government than we have today to address this problem. We don’t seem to have the ability needed for long-term planning.”

“This is an exciting time, as we learn more and more about free-flowing rivers and the good they do for communities and for the environment.” –Evan Dethier

Along those lines, Best says, “transnational agreements are critical. Large rivers are often transboundary rivers, so we can’t do anything without a proper governmental and management structure.” He points to the Danube and Rhine rivers, which together cross 14 countries. “In the last 20 years, there’s been much better agreement between those nations about protecting the rivers and their environment.”

While Best sees promise in the future health of rivers where dams are being decommissioned and through technologies such as NASA’s Surface Water and Ocean Topography (SWOT) mission that allow for better remote monitoring of river flows, he says that dams and reservoirs being built or planned in the Global South are “where the new pressures are the most extreme.” He cautions: “We need to learn from the things we haven’t done well in the last 50 years.”

Dethier, now an assistant professor at Occidental College in Los Angeles, emphasizes Best’s point, saying, the next decade will be critical, “and not just in rich nations. All environmental challenges are interconnected. We can, and we must, do a better job of taking care of the environment.” He continues on an optimistic note: “This is an exciting time, as we learn more and more about free-flowing rivers and the good they do for communities and for the environment.”

In Desert Solitaire, Edward Abbey later describes what one might see if such free-flowing rivers went on without an end, through pastoral valleys: “something strange and more beautiful and more full of wonder than your deepest dreams waits for you — beyond that next turning of the canyon walls.”

The post Why it’s important to let sediment reach the shore appeared first on Ensia.

Early contraceptive discontinuation and associated factors among married women initiating long-acting and short-acting contraceptives in humanitarian settings in Ethiopia: A retrospective cohort study

Juliana Yartey Enos · PLOS ONE Feed

by Andamlak Gizaw Alamdo, Agnes M. Kotoh, Emefa Judith Modey, Juliana Yartey Enos

methods, especially among urban refugee women, are crucial.

The sexual and reproductive health (SRH) service provision in conflict-affected populations has been continually overlooked. International and local relief organizations have begun to recognize the neglect of women's SRH needs; however, SRH services and humanitarian responses remain suboptimal. Contraceptive discontinuation is a significant public health issue with considerable impact on family planning (FP) programs, as well as on women's SRH and population growth. There is scarce information regarding the dynamics of contraceptive use in refugee areas. Evidence on contraceptive use dynamics is vital, especially now, as the global refugee crisis grows. Therefore, this study aimed to examine first-year contraceptive discontinuation and associated factors among married women initiating long-acting reversible contraceptives (LARC) and short-acting contraceptives (SAC) in selected refugee populations in Ethiopia.

A retrospective cohort study was conducted among 551 married women in selected refugee populations in Ethiopia. After reviewing FP registries, all eligible women who initiated a contraceptive method 12–18 months before data collection were included in the study. Through computer-assisted face-to-face interviews, data on the socio-demographic, reproductive, and contraceptive use history of women were collected. Key characteristics of SAC acceptors and LARC acceptors were compared using chi-square statistics and t-tests. A multivariable Cox proportional hazards model identified the predictors of time to contraceptive discontinuation during the first year of use. In addition, adjusted hazard ratios (aHR), 95% CI, and P-values were calculated to determine the magnitude and precision of the association.

The 12-month discontinuation rate for all contraceptive methods was 41.2% (95% CI: 37.2–45.4). The discontinuation rate for SAC (44.94%) was higher than LARC (32.32%) ($P \leq 0.001$). The factors associated with 12-month discontinuation include method type (aHR = 1.50; 95% CI: 1.06–2.12), residence (aHR = 1.83; 95% CI: 1.24–2.70), husband occupation (aHR = 0.49; 95% CI: 0.29–0.84), length of stay in the refugee camp (aHR = 0.95; 95% CI: 0.92–0.99), and intention to use contraceptives in the future (aHR = 0.15; 95% CI: 0.11–0.21).

A high rate of early contraceptive discontinuation was documented among refugee women, with a higher discontinuation rate among SAC acceptors. To improve contraceptive service provision and maintain continuous use of the methods initiated, targeted interventions, such as improved counselling and promotion of long-acting contraceptive

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FEATURES

Apple issues urgent lock screen warnings for unpatched iPhones and iPads

Pierluigi Paganini · Security Affairs

Apple is sending lock screen alerts to users running outdated iOS and iPadOS versions, warning of active web-based attacks targeting their devices. The notifications urge users to install critical updates to stay protected, highlighting ongoing exploitation of older software versions.

“The alerts, which appear as a “Critical Software” notification from the Settings app, warn that Apple “is aware of attacks targeting out-of-date iOS software, including the version on your iPhone,” and urge users to install a critical update to protect their device.” reported MacRumors .”The notifications are being seen on devices running a range of older iOS versions, including iOS 17.0, far beyond the iOS 13 and iOS 14 devices that Apple specifically flagged in its support documentation .”

Apple warns that exploit kits like “ Coruna ” and “ DarkSword ” are actively targeting iOS versions 13 through 17.2.1. The company warns that on unpatched devices, simply clicking a malicious link or visiting a compromised website could allow attackers to exploit vulnerabilities and steal sensitive data.

“If your iPhone doesn’t have the latest software, update iOS to protect your data,” states Apple.

Coruna targets iOS versions 13.0 to 17.2.1, while DarkSword is aimed at newer devices running iOS 18.4 to 18.7.

Recently, Apple warned that iPhones running outdated iOS versions are at risk from exploit kits like Coruna and DarkSword . These attacks use malicious web content to trigger infection chains that can steal sensitive data. Users are strongly advised to update their devices to stay protected.

“Security researchers recently identified web-based attacks that target out-of-date versions of iOS through malicious web content. For example, if you’re using an older version of iOS and were to click a malicious link or visit a compromised website, the data on your iPhone might be at risk of being stolen.” reads Apple’s advisory . “We thoroughly investigated these issues as they were found and released software updates as quickly as possible for the most recent operating system versions to address vulnerabilities and disrupt such attacks.”

Keeping the iPhone updated is the most effective way to stay protected from threats like Coruna and DarkSword. Devices running the latest iOS versions are not vulnerable, and Lockdown Mode also blocks these attacks, even on older systems, though updates are still strongly recommended.

If your iPhone runs an older iOS version, take action:

Devices on iOS 15 to iOS 26 are already protected if fully updated

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Apple released updates on March 11, 2026, to extend protection to iOS 15 and 16 devices

Devices on iOS 13 or 14 must upgrade to iOS 15 and install a Critical Security Update

Safari’s Safe Browsing feature helps block known malicious domains by default

Updating ensures user data remains secure.

This week, Kaspersky researchers reported that the Coruna iOS exploit kit uses an updated version of the same kernel exploit seen in the 2023 Operation Triangulation campaign. While early evidence didn’t clearly link the two, the code similarities now suggest a possible connection between them, though shared vulnerabilities alone don’t definitively prove the same actors are behind both attacks.

Researchers collected and analyzed Coruna components, confirming strong code similarities. The kit also includes four additional kernel exploits, some developed after Triangulation , all built on the same framework.

These findings suggest Coruna is not a mix of reused parts but a more advanced evolution of the same exploitation framework behind Operation Triangulation.

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(SecurityAffairs – hacking, Coruna)

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Embracing Afghanistan's refugees into the UK's health and social care system

julietwalker · The BMJ Blog

The response to refugees arriving in the UK after the international emergency in Afghanistan requires health professionals to be equipped for the needs of this population. The integration of this community from a health and social care perspective is key in terms of ensuring equitable provision. The Equality and Diversity Act (2010) must frame this planning, from where services can monitor and embrace this community to ensure they are not being disproportionately impacted. [1] The UK has committed to accepting 20,000 refugees over five years in response to the political takeover by the Taliban in Afghanistan. Healthcare professionals will need to develop allyship and effective working practices including challenging discrimination, stereotypes, racism, and xenophobia.

There must be relevant training and resources within health and social care teams and departments to ensure that refugees' needs are met and communication is safe. The Cultural Atlas (2021) provides one of the most comprehensive educational tools in understanding the culture and guiding healthcare practice on etiquette and communication. [2] The lack of available acceptable housing is going to be one of the largest issues facing Afghan refugees. There needs to be a close working relationship between community health services, homeless teams, and housing services. Such relationships are key as there is a duty of care to anyone in the local area to be housed even if that is in temporary accommodation. Health professionals may be the first responder in ensuring that local signposting to organisations and advocacy agencies are appropriately done.

Another issue that will need to be addressed in a timely fashion is the lack of money and finances. Contrary to myths surrounding refugees and asylum seekers who enter the country, they will not have money to live the lifestyle alleged by many. An asylum decision takes at least 28 days and at this point, there may be more destabilisation incurred for the families as temporary accommodation could cease. The usual expectations are to secure work or study but often children of this group barely get food, clothes, or toys, and families rely heavily on charitable donations.

Telephone language services have been a long-term resource available in the UK. The language needs of the Afghan community are likely to be Pashto and Dari. In areas of high dispersal, the use of key workers from the same background would be useful in ensuring others in teams learn about the cultural and religious disposition of this community.

There may be a need to refer or to seek advice from safeguarding teams, where a child is in need or at risk. We cannot assume that the refugee status mitigates such referrals, and any impact on children is a usual way of life. Where there are needs not being met in the form of finances or a child is not able to get to school, for example, social

services have a duty of care to ensure both risk and "needs" are assessed under the Children's Act.

Trauma interventions in terms of what is an effect of resettlement, or witnessing events in Afghanistan recently or in the past, will require robust resources. Combat stress, a veteran support line, reported an exceedingly high number of calls from veterans who served in Afghanistan and required mental health support, including suicidal ideation due to post traumatic stress disorder. [3] Prioritising support within the NHS workforce, which has already been impacted by covid-19 will be quite a challenge. Additionally, services need to be aware that culturally many Afghan refugees will feel more at ease talking in a group setting, as opposed to individual support sessions. Grief is often seen as a community experience rather than an individual one. The need to develop community kinships and innovative methods to engage with these groups would be advisable.

Suffering, grief, and ambiguous loss will feature in many presentations, which will not be restricted to trauma. Separation from loved ones and death related losses will be part of the psychological needs of the refugees. Therefore, there may be a need to increase the service provision in primary care psychological services for trauma and complex grief, at local and national levels.

Health services and the health and social care workforce will need to be skilled and culturally ready to welcome Afghan refugees to the UK. The double burden of covid-19 and poor healthcare has impacted the already burdened healthcare system in Afghanistan. [4] After ensuring the population is registered with a local GP, another complicating factor in the present circumstances is the impact of covid-19. Health professionals will have to quickly establish the refugees' covid status, where they are in the vaccination programme, and introduce the refugees to covid-19 adjustments that the UK population has become used to. A big factor impacting the refugees will be their housing. If, as seems highly likely, they will be housed in overcrowded accommodation then this is the perfect breeding ground for covid-19 to spread. As stated, many will arrive in the UK in physically poor health, that in combination with substandard accommodation, and the threat of covid may prove fatal. Health professionals must be outspoken in trying to stop this scenario from unfolding.

The health and social care interface have the resources to support and ensure that a good value base is demonstrated through their practice. The disproportionate impact of covid-19 has already affected ethnic minorities across the globe. There is a small window of opportunity for services now to predict and assess their cultural responsiveness rather than rely solely on less appraised community-based initiatives.

Harjinder Kaur-Aujla, lecturer in mental health nursing, University of Birmingham.

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Inundación E Injusticia: Las Inundaciones Representan Una Tremenda Amenaza Para La Región De Los Grandes Lagos

Kari Lydersen · Ensia

Nota del editor: Este reportaje es la introducción de una colaboración en seis partes — “Inundados: Inundaciones y comunidades vulnerables en la región de los Grandes Lagos” — que incluye al Institute for Nonprofit News, Borderless, Ensia, Grist, Planet Detroit, Sahan Journal y Wisconsin Watch, así como a The Guardian e Inside Climate News. El proyecto contó con el apoyo de la Fundación Joyce. Para leer la versión en inglés, haga clic aquí .

El fin de semana del Día de la Independencia de este verano iba a ser un momento histórico para Chicago, con la celebración de la primera carrera de NASCAR con carros circulando a toda velocidad por el centro de la ciudad. Pero estos planes se desbarataron parcialmente, ya que se hizo historia por otra razón.

Un récord de lluvias inundaron las calles de la ciudad, sumergieron pasos subterráneos y más de 2,000 sótanos, muchos de ellos con aguas residuales. Como muchas catástrofes climáticas, las inundaciones afectaron de forma desproporcionada a los más vulnerables de la ciudad, como los inmigrantes y las comunidades de color.

La aumento de El caudal del río Chicago se invirtió las aguas pluviales y residuales sin tratar se vierten en el lago Michigan, fuente de agua potable para millones de personas. El alcalde Brandon Johnson, que vive en la zona oeste de la ciudad, muy afectada por las inundaciones, describió el desastre como un signo del cambio climático y un presagio de lo que está por venir.

Unas precipitaciones récord inundaron Chicago durante el fin de semana del Día de la Independencia de este año. Imagen por cortesía de la Administración Nacional Oceánica y Atmosférica, Servicio Meteorológico Nacional

Como esta colaboración de seis partes va a resaltar, las ciudades de toda la región de los Grandes Lagos se enfrentan a crisis similares, con sistemas de aguas residuales arcaicos, infraestructuras en ruinas y viviendas segregadas que crean una tormenta perfecta de vulnerabilidad a las inundaciones de lugares que van desde el exceso de lluvia y el desbordamiento de los ríos hasta las mareas de tempestad lacustres y las inundaciones de los sistemas de alcantarillado. Las zonas rurales, las comunidades indígenas y los ecosistemas de los Grandes Lagos también se enfrentan a graves riesgos de inundación, lo que pone en peligro los logros conseguidos con esfuerzo en materia de restauración ambiental y desarrollo comunitario. En todos los casos, los problemas de justicia ambiental van de la mano de los riesgos de inundación, y la colaboración con las comunidades afectadas es clave para encontrar soluciones.

En junio, la administración Biden anunció importantes iniciativas de resistencia al cambio climático que

incluyen la reducción del riesgo de inundaciones, el “apoyo y aprendizaje de las comunidades tribales” y la promoción de la justicia ambiental.

La Ley Bipartidista de Infraestructuras y la Ley de Reducción de la Inflación incluyen financiación para hacer frente al cambio climático y reforzar la capacidad de recuperación de las zonas costeras, incluidos los Grandes Lagos. La Ley de Reducción de la Inflación incluye un proyecto de 575 millones de dólares de la Administración Nacional Oceánica y Atmosférica (NOAA) para ayudar a estas comunidades con “infraestructuras naturales”, llevando a cabo una “reubicación dirigida por la comunidad” y protegiéndolas de otro modo contra los impactos climáticos extremos, en consonancia con la Iniciativa de Justicia40 de la administración que ordena que al menos el 40% de las inversiones se destinen a comunidades desfavorecidas.

La Ley Bipartidista de Infraestructuras también incluye 11,700 millones de dólares — además de los fondos básicos existentes — para el Fondo Estatal de Renovación de Agua Limpia (Clean Water State Revolving Fund) que ofrece préstamos inferiores a los del mercado y algunas subvenciones federales para mejorar las infraestructuras de tratamiento de aguas residuales. Esto es especialmente importante para hacer frente a los sistemas combinados de alcantarillado que transportan tanto las aguas pluviales como las residuales y se desbordan durante las lluvias torrenciales, haciendo que las aguas residuales burbujeen hasta los sótanos y obligando a verter las aguas residuales tanto ríos como en lagos.

Las mejoras de las infraestructuras municipales de gestión del agua que separan los colectores de aguas pluviales de los sanitarios tienen como objetivo reducir o eliminar el vertido de aguas residuales sin depurar a las aguas superficiales durante fenómenos meteorológicos intensos. Cortesía del Distrito Metropolitano de Alcantarillado de Milwaukee

Las inundaciones de sótanos provocadas por sistemas de alcantarillado sobrecargados no están cubiertas por el Programa Nacional de Seguros contra Inundaciones (NFIP), administrado por la Agencia Federal para la Gestión de Emergencias (FEMA), que proporciona seguros contra inundaciones asequibles, elabora mapas de riesgo de inundación y obliga a cumplir las normas de zonificación y construcción relacionadas con las inundaciones en las casi 23,000 comunidades cubiertas por el programa.

“Sería poco realista esperar que la FEMA trazara mapas de las zonas vulnerables [del sistema de alcantarillado] a las inundaciones urbanas porque tendría que disponer de información perfecta sobre los sistemas de alcantarillado [de las ciudades] y, francamente, la mayoría de las ciudades no disponen de esa información”, afirma Rob Moore, analista político del Consejo de Defensa de los Recursos Naturales (NRDC).

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Covid-19 vaccination in children, adolescents, and young adults: how can we ensure high vaccination uptake?

julietwalker · The BMJ Blog

After a rapid start, the pace of the United Kingdom's (UK) covid-19 vaccination programme has slowed down while the UK still faces high infection, hospitalisation, and death rates, and a more transmissible Delta SARS-CoV-2 variant. Now that vaccination of children aged 12-15 has started, it is essential to achieve a high uptake of vaccination in this group, and also in young adults, to both protect them and to move the UK closer towards population level immunity. [1,2] Despite two doses of Pfizer-BioNTech, Moderna, and AstraZeneca vaccines offering good protection against the Delta variant— with Pfizer-BioNTech and AstraZeneca vaccines between 92-96% effective in preventing hospitalisations— many young people remain unvaccinated by choice, raising their risk of infection, hospitalisation, and long-term complications from covid-19. [3-5]

The UK population is among the most willing to receive a covid-19 vaccine; as of 11 October 2021, over 49 million individuals (85.6% of people aged 16 and over) had received at least one dose of a covid-19 vaccine. [6, 7] However, the covid-19 vaccination programme—the largest ever launched by the NHS—is reaching a “demand” ceiling in adolescents and young adults, finding itself well behind other Western European countries, and hampering efforts to achieve population level immunity. If vaccination uptake is also slow in 12-15 years old children, this will further hinder efforts to reach population immunity.

Vaccination rates in younger people are lower and increasing more slowly than was seen in older age groups when they were first offered vaccination. [1,8] According to the Office for National Statistics, 14% of those aged 16-17 years, 10% of those aged between 22-25 years, and 9% of those aged between 18-21 years consider themselves “hesitant” compared to 4% observed across all other age groups. [5] This mirrors concerning findings from the USA which demonstrate that one in four of those aged between 18 and 25 “probably will not” or “definitely will not” receive a covid-19 vaccine, despite their heightened infection risk in recent months. [9] Given their increased tendency to socialise, strategies that improve vaccine acceptance in adolescents and young adults remain essential to control the pandemic globally as well as in the UK. [10]

Historically, vaccine hesitancy exists on a spectrum and is listed by the WHO as one of the top 10 global health threats. [11] The groups that are among the currently most affected by the virus are also the ones with the lowest vaccination rates. [1 2] With ideal conditions for SARS-CoV-2 to spread, the risk of emergence of “super variants” that could potentially escape vaccines and jeopardise the health of the most vulnerable in society remains a risk. Vaccine hesitancy in young people in the UK may be further increased by the delay in approving vaccination for 12–15 year-olds, with the UK starting vaccination later than many other European and North American countries. The message from the UK's

Joint Committee on Vaccination and Immunisation (JCVI) that covid-19 vaccination in this group offers only “marginal benefits” will also have contributed to this, with many parents and children questioning why they should be vaccinated if this is the case. [13] The benefits and potential risk from vaccination will therefore need to be discussed carefully with children and their parents to dispel any unwarranted negative views. [14]

This has been successfully done in Portugal; despite Portuguese parents not being safe from vaccine misinformation and disinformation, the country has managed to emerge as the world's vaccination front-runner, with 86% of its population vaccinated (98% of whom are aged 12 years and over). [15] Its successful vaccine rollout is, in part, attributed to the country's comprehensive monitoring system; vaccine compliance is monitored nationally by healthcare facilities, schools, daycare centres, summer camps, and other child institutions, allowing the country to develop and tailor educational information to hesitant parents or parents known to have refused a vaccine in the past. [16] This has generated favourable conditions for paediatric immunisation across the country.

Concerns about side effects are an important factor in vaccine hesitancy in children, adolescents, and young adults, particularly the risk of condition such as myocarditis. [9] Although rare, the myocarditis and pericarditis reports in adolescents and young adults, following the administration of Pfizer-BioNTech and Moderna vaccines, will have amplified fears of vaccines in this group. [17] However, the risk of developing complications, such as blood clots and myocarditis, from covid-19 illness remains greater than the risk from vaccines. [18] Genuine concerns about the side effects of vaccines should be addressed by academics and clinicians proactively listening to young people, and sharing risks and benefits in a manner that aligns intention with action. [19] It is also essential that moving forwards, the UK's covid-19 vaccination programme is embedded in primary care to create a cost-effective, sustainable infrastructure for vaccine delivery; and to avoid making the many mistakes that were made in other parts of the covid-19 response, such as Test and Trace and the Nightingale Hospitals. [20]

To offset optimistic bias, including adolescents and young adults perceiving the risk of disease being lower than the risk of receiving a covid-19 vaccine, communication should speak to mechanism of action, effectiveness, and safety relevant to these age groups and the wider societal benefits of vaccination in protecting their older family members, and vulnerable friends and colleagues. [10,21] Further, public health messaging will be more effective if the benefits of controlling the pandemic, including freedom to attend festivals, sporting events and entertainment venues, as well as the ability to travel are reinforced. Targeted health messaging and public education campaigns will also require harnessing social media, schools and universities to counter the covid-19 infodemic. [10] To increase vaccination

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Turning the tide: The Obesity Health Alliance's healthy weight strategy

julietwalker · The BMJ Blog

Tides are notoriously difficult to turn. But, when the time is right, even the strongest tides will turn—and now is one of those critical moments in relation to overweight and obesity in the UK. This follows 30 years of piecemeal action by governments to address a preventable health crisis that has become even more apparent during the covid-19 pandemic. Covid-19 has shone a light on the underlying health problems that make many of us more vulnerable when a new disease threatens population health and health systems that are already overstretched.

Recently the Obesity Health Alliance (OHA)—a coalition of 45 leading health charities and medical royal colleges—launched their new report “Turning the Tide: A 10-Year Healthy Weight Strategy” for the UK. Obesity rates have increased significantly across the UK over the last 30 years, despite 14 government health strategies since 1991 that have set targets for obesity reduction in England, with similar failed efforts in the devolved nations. In total, these strategies contained 689 recommendations. Today, the majority of adults in England—68% of men and 60% of women—are above a healthy weight, and over a quarter are living with obesity (27% of men and 29% of women). Excess weight prevalence is highest among the lowest socioeconomic groups in both adults and children.

The OHA's strategy was developed over two years by a working group chaired by Anne Johnson. Through a process of consensus-building, informed by a series of evidence reviews and expert papers, the strategy includes 30 recommendations for action in the years ahead.

It notes that the commercial food system has fundamentally changed over the second quarter of the 20th century. The key goal of this food system was to eradicate hunger and ensure adequate food supply, however it has led to a complex adaptive system in which the need for readily available and accessible food has driven an explosion of ultra-processed foods that are highly palatable and heavily marketed to both meet and drive consumer demand. This has altered what we eat and how we behave in the same way that commercially mass produced cigarettes made smoking a normal part of life for the vast majority last century. Smoking resulted in hundreds of thousands of preventable premature deaths in the decades that followed until governments took action and changed the system to prevent smoking uptake and support existing smokers to quit.

The evidence is clear that the drivers of healthy weight are even more complex than those involved in tobacco use. The obesogenic environment, to which we are all exposed to from infancy onwards, whereby calorie-dense, nutrient poor food is readily accessible, abundant, affordable, and normalised, and physical activity opportunities are not easily available, affects us all. The pervasive and ubiquitous advertising and promotion of unhealthy food and drinks allows industry to “set the tone” of our food environment,

creating, and maintaining this obesogenic environment that makes it inherently difficult to achieve a healthy diet.

Obesity cannot be addressed one person at a time, and behind the statistics are people living with obesity who often experience stigma and discrimination that can affect their mental health and willingness to access healthcare. Stigma stands in the way of public health and must be eradicated.

The strategy makes recommendations across key areas: 1) stigma; 2) food and drink products; 3) the marketing mix; 4) pricing; 5) the environment; 6) advertising and promotions; 7) early years; 8) management, treatment and support; and 9) improved policymaking processes. Each of these areas require addressing in their own right, but combined the recommendations set the clear, long-term, evidence-informed agenda needed to turn the tide and improve healthy weight across the UK population. The 30 recommendations made are set out in a “KIND” framework, aimed at building on existing policy progress as well as identifying new routes for action. The KIND framework includes:

Keep policies already in place or that are due to be implemented that supports a healthy weight environment;

Intensify existing policies or approaches to increase impact;

New proposals that are recommended for evidence-informed actions;

Develop policies based on the results of new, promising areas for research and investment.

The framework sets out not only the stage of policy development in which each recommendation sits, but also the actor(s) that should enact that recommendation. It is not enough to simply name route for action, but we need to establish clear directives for those bodies and organisations responsible to implement necessary change.

The role of industry is not absent here, and it is vitally important that we acknowledge the power that companies currently has over our food environment, as well as the regulation of the environment. We have clear evidence demonstrating that industry involvement in policy-making designed to remedy the harms industry products contribute to is not conducive to the creation of healthy public policy, and this must be actively managed throughout the policy process.

As we look ahead to recovery from the covid-19 pandemic, progressing efforts to achieve a healthy weight across the UK population must be at the heart of public health. It is time to move beyond individual-level policy recommendations, and implement wide-ranging changes to the system that got us here in the first place. This will not only save lives in the short and medium term, but will also serve to create a better and healthier future for our children and grandchildren, something they most surely deserve.

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We must not delay speaking up about the health harms of the climate emergency

julietwalker · The BMJ Blog

In paediatric medicine, denial of impending disaster prevents us from recognising when a child's medical condition is deteriorating. Someone uncertain of their role, or afraid to speak up because of rigid hierarchies, may choose to say nothing. Delay in starting the right treatment may be caused by fear of appearing to over react. We instinctively wish to find a simple technical cause for an alarm, with a simple technical fix. We are trained in this bad habit by incessant alarms, while they initially sound worrying, they usually turn out to be false and we hit mute. Thus, we may miss the fact that the oximeter cannot get a signal due to hypoperfusion from septic shock and impending cardiovascular collapse.

In much the same way humanity and the medical profession have denied our role in human-caused global heating and the destruction of nature for several decades. Now that the vast majority of the world's population recognises that the anthropogenic climate health emergency is happening, those still unconvinced or unable to act have pivoted from denial to delay. The correct prescription for climate action is well known and universally accepted by science—the immediate and just transition from fossil fuels to renewables, the reversal of destruction of our natural habitats and restoration of biodiversity in nature— we still focus on unproven and potentially unscalable technological fixes such as carbon capture. What is more, the success of victim blaming that has occurred through the promotion of personal carbon footprinting has successfully diverted attention away from the correct treatment which is to stop extracting and burning coal, oil, and gas . Just to make certain that no one puts their head above the parapet, the ubiquitous use of personal responsibility , has become deeply entrenched in discussions around the climate health emergency. It would be unusual to have a discussion about the climate without being asked if you still eat beef or drive a diesel car (open disclosure my answer is; occasionally and no).

So, because the adults have left the room, the burden is placed on children and the younger generation to do the advocacy, to miss school, and to be trolled by the political leaders of the free world.

But as healthcare professionals we can understand science. We have experienced the impact that the tobacco industry and “big pharma” can have with their vast financial and lobbying resources when they choose profit over the survival of their customers. We would (now) never consider taking bribes from big tobacco to pretend that cigarette smoke doesn't cause lung cancer. But what have we said about the same particulates that come from our car exhausts and wood burning stoves? Not nearly enough according to one coroner .

It is time we stopped worrying about our inadequate skills and knowledge in the science of air quality and the climate

health emergency. We don't have time to become experts in climate science before speaking up about the health harms of the climate emergency. Whenever you are told that this is not in your job description and what you are saying is political, look at their motives for saying that. Are they in children's best interests or the status quo?

The Climate Health Emergency is the greatest threat to child health in human history. We need to declare an emergency and then act as if there is an emergency, not sit back and wait for someone else to take charge. We need to use the skills we have at once.

If we work as a team and use all our individual strengths to achieve a common goal, we can give “the climate policy ratchet a healthy turn .”

We need team leaders who;

style="font-weight: 400;"> Name and clearly articulate goals

style="font-weight: 400;"> Set specific targets and the methods to achieve them

style="font-weight: 400;"> Know their own and our team's strengths and weaknesses.

And other roles are equally important:

style="font-weight: 400;"> “Finishers” who see errors and fix them

style="font-weight: 400;"> “Coordinators” who see opportunities and allocate resources.

style="font-weight: 400;"> “Evaluators” who can stand back and prevent task fixation errors.

Teams can achieve major feats that individuals cannot.

But they need leadership, cooperation, coordination, and communication. Nothing will happen until we all speak the truth. Denial, and its best friend, delay, do not lead to good outcomes.

Mark Hayden is a paediatric intensivist working in London. He is taking part in the Ride for Their Lives from London to Glasgow ahead of COP26. He knows that this action will not reduce the concentration of carbon dioxide in the atmosphere, but hopes that this action will inspire other actions that will.

Competing interests : none declared.

The post We must not delay speaking up about the health harms of the climate emergency appeared first on The BMJ .

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Milwaukee residents fear more flooding due to planned highway expansion

Jonah Chester · Ensia

Editor's note: This story is part of a collaboration that includes the Institute for Nonprofit News, Borderless, Ensia, Planet Detroit, Sahan Journal, and Wisconsin Watch, as well as the Guardian and Inside Climate News. The project was supported by the Joyce Foundation. You can read the launch story from Ensia, "Inundation and Injustice: Flooding presents a formidable threat to the Great Lakes region," here. This story was originally published by Wisconsin Watch, a nonpartisan, nonprofit investigative news outlet reporting on issues of importance to Wisconsin.

[Click here for highlights from the story.](#)

The Wisconsin Department of Transportation plans to spend about \$1.2 billion to add two lanes to a 3.5-mile segment of the I-94 East-West Freeway corridor in Milwaukee, aiming to repair the road and ease congestion.

Residents worry the addition of about 29 acres (12 hectares) of asphalt will increase flows of stormwater into surrounding flood-vulnerable neighborhoods.

The Wisconsin Department of Transportation acknowledges the extra pavement will increase stormwater runoff. The department doesn't plan to analyze precise effects on runoff until the final design phases of a project expected to break ground in 2025.

More than 20% of households near the corridor lack a car, and expansion opponents point to volumes of research showing that widening highways can actually increase traffic volume.

Janet Haas two decades ago saw potential in a field of thistle, grass and bushes that Milwaukee County had neglected: Valley Park, nestled between the Menomonee River to the west and homes in one of Milwaukee's most racially diverse neighborhoods to the east.

"The county doesn't have any money, and they haven't had any money for years," says Haas, who has lived most of her life in the Near West Side neighborhood, called The Valley or Pigsville. "One day I just decided that the thistles were as tall as I was, and I wasn't going to take it anymore."

That meant persuading her mom and a friend to join her in pulling weeds and planting bushes and flowers in the park. The Milwaukee Metropolitan Sewerage District built the park as part of a multi-million dollar project to protect the neighborhood from flooding – but, Haas says, it fell into disrepair after its completion in 2001.

Valley Park is now a meticulously maintained community gathering spot – complete with walking and biking trails, a play structure and shade trees. Residents take pride in the tidy park – evidenced by their weekly cleanup events, such as a June 22nd gathering of about a dozen neighbors.

"We all know each other, we look out for each other," says Moses Mcknight, a resident of 17 years, as children and adults moved mulch and watered flowers in the park. "We like to fix things up, and keep it that way."

But neighbors at the Thursday evening cleanup worry that a US\$1.2 billion plan to widen the Interstate 94 East-West Freeway corridor will disrupt their gatherings and undo progress.

I-94's six lanes stretch above the Menomonee River and run south of Valley Park — carrying commuters between downtown Milwaukee and the city's western suburbs in Waukesha County. Backed by Democratic Gov. Tony Evers, the Wisconsin Department of Transportation plans to add two lanes to a 3.5-mile segment of the highway.

A view of Milwaukee's Valley Park looking southwest towards American Family Field. Local residents worry that a US\$1.2 billion plan to widen the Interstate 94 East-West Freeway corridor will undo progress in beautifying the park. Photo by Pat A. Robinson for Wisconsin Watch

Valley Park sits in the middle of the planned expansion. Aside from increasing noise and air pollution as more cars zip down a wider highway, residents worry the addition of about 29 acres (12 hectares) of asphalt — the equivalent of more than twenty football fields — will increase flows of stormwater into The Valley and surrounding flood-vulnerable neighborhoods.

The state transportation department acknowledges the extra pavement will increase stormwater runoff, but it says it won't analyze precise effects on runoff until the final design phases of a project expected to break ground in 2025.

"Who's going to benefit? Because it's not going to be us," Haas says. "The soil is going to be crummier and more polluted. We worked on this park, why should we give it back?"

The transportation department declined to be interviewed for this story. In an unsigned statement emailed by a spokesperson, the department wrote, in part: "We work to avoid or minimize the impacts of infrastructure improvements to the natural and human environment while delivering projects efficiently."

Evers and the Wisconsin Department of Transportation want to expand the interstate to decrease traffic congestion and high crash rates, and repair aging infrastructure in what Evers has called one of the state's "most congested and dangerous roads."

Menomonee Valley Partners, a nonprofit dedicated to redeveloping the Menomonee River Valley, has praised the state's decision to proceed with the project.

On the right is a map showing census

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Can state-level green amendments protect from rollbacks in federal regulations?

Brian Allnutt · Ensia

A proposed state constitutional amendment introduced in the Michigan House in April would put the right to clean air, water, soil and a stable climate on par with other fundamental liberties like the freedoms of speech and religion.

The Michigan Green Amendment, introduced by Rep. Rachel Hood (D-Grand Rapids), follows the passage of amendments in Pennsylvania, Montana and New York that have been used to fight environmental threats like PFAS contamination and hydraulic fracking in residential neighborhoods. More than a dozen other states are also considering such amendments.

“The goal is to make sure that our government is making better decisions so that our fundamental rights are protected,” says Maya K. van Rossum, founder of the nonprofit Green Amendments For the Generations. The amendment could offer legal recourse for residents whose right to a healthy environment is infringed upon and protect state laws from federal environmental rollbacks.

Pennsylvania’s green amendment has helped environmental advocates secure important victories in recent years. Although the state’s Environmental Rights Amendment was enacted in 1971, it didn’t have much impact until 2013, when the Pennsylvania Supreme Court cited the amendment in its decision to overturn a requirement that municipalities allow oil and gas developments in all zoning areas.

In 2018, former Democratic Pennsylvania Governor Tom Wolf referred to the amendment in an executive order creating the state’s PFAS action team, which was followed by the state setting limits on two common types of PFAS in drinking water.

Pursuing green amendments at the state level could be far easier to pass than a federal constitutional amendment and win substantial protections for residents, van Rossum says.

“When it comes to environmental protection, states have a tremendous amount of authority,” she says. Indeed, on issues like regulating PFAS chemicals and addressing vehicle emissions, states often lead the way in adopting stricter environmental protections.

Green amendment advocates argue that the approach establishes a framework integrated into state policymaking, protecting residents from environmental threats and preventing problems before they arise, as opposed to passing laws against individual environmental threats.

Washington state Rep. Debra Lekanoff (D) previously said an amendment would ensure that “every environmental policy that you build [in] Washington state, every regulation, everything you build across the state, every fiscal note you invest in Washington state is adequately providing the healthy environment.”

However, some have argued that green amendments could lead to litigation that blocks essential projects. The New Mexico Legislature’s Legislative Finance Committee said in a 2023 report that a green amendment could potentially be used as a “roadblock to pursuing clean energy projects as part of New Mexico’s renewable energy transition,” and that, “the legal uncertainty the amendment could create might result in costly litigation that could impact the financial feasibility of certain energy projects.”

In response, van Rossum said New Mexico’s proposed amendment could actually help clean energy projects defend against legal challenges because of the clause requiring the state to safeguard a stable climate.

She has also argued that green amendments could protect states from rollbacks in federal regulations. The prospect of a second Trump presidency could give this argument more urgency as the Heritage Foundation and other conservative organizations advocate for shrinking the Environmental Protection Agency and directing its focus away from the climate crisis.

States frequently pass constitutional amendments. The last amendment to the U. S. Constitution was passed in 1992, whereas Michigan passed three in 2022 alone.

In Michigan, a constitutional amendment can be passed either by a two-thirds vote of both chambers of the legislature, at a constitutional convention or with a ballot initiative.

Passing such a measure with a ballot initiative may be more likely given the closely divided legislature.

But van Rossum says that, for now, the important thing was to educate lawmakers and voters about what such an amendment would mean, which could also lead to a push for a federal amendment.

“Going state by state by state gets us critical protections in the near term and also allows us to undertake the education and the organizing needed to ultimately succeed at the federal level,” she says.

Editor’s note: This article first appeared on Planet Detroit and is republished here under a Creative Commons license. Ensia and Planet Detroit previously collaborated on a reporting project, “Inundated: Flooding and vulnerable communities in the Great Lakes region,” along with the Institute for Nonprofit News, Borderless, Sahan Journal, Wisconsin Watch, the Guardian and Inside Climate News, which looked at flooding vulnerability for cities, rural areas and Indigenous communities throughout the Great Lakes region from sources that range from excessive rain and overflowing rivers to lake storm surges and sewage system flooding.

The post Can state-level green amendments protect from rollbacks in federal regulations? appeared first on Ensia .

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Apple Jelly

David · David Lebovitz

I was recently reunited with something I miss very much – a loaded apple tree! Friends of mine who had a house in the French countryside had a tree that, come fall, had so many apples, the limbs threatened to break off. Not wanting to be an accomplice in apple-cide, I decided to do my part to save the tree, and the apples, and make Apple Jelly.

Ten years later, my friends sold their house, which ended my bounty of apples (and medlars .) So how happy was I when some other friends bought a house in Brittany, which had not one, not two...but four apple trees. Score!

The varieties were Reine de Reinette, Canada, and Boskop, which I found out when a concerned neighbor saw someone (me) swiping apples off the trees, and the ground, loading up sacks of them. I think he figured I was pilfering the pommes , but was reassured when I told him I was friends with the owner of the mini-orchard.

After talking with him a few minutes, I realized he was an expert on the apples and not only knew about the varieties of apples, but also what to do with them and nodded in approval when I told him I was going to make Apple Jelly. He was happy to help me fill some bags, which included green varieties, which are high in pectin and help the jelly set, and are more acidic than other apples, which make the jelly well-balanced and delicious. (Calvados, the famed apple brandy from Normandy, use very acidic apples since the flavor is more pronounced once cooked.)

To make Apple Jelly, it's probably a good idea to either a friend with an apple tree (with an explanation ready in case a neighbor comes by...) or get apples from a source where they might sell you apples that are dented or dinged, which are called pommes à cuire (cooking apples) in France, and are quite popular. At the markets, those apples are usually the first to get sold as people use the bargain beauties to make compotes and tarts with. And French bakers like to use a variety of apples in dessert, like Apple Cake , as the flavor is more interesting than if using just one variety.

In the end, I came home with over 50 pounds (23kg) of apples in all shapes, colors, and conditions. And all went to a good cause. This recipe uses a good deal of apples but makes a half-dozen jars, which is good if you're anything like me and need to reward your friends, and perhaps a local apple expert, with a jar of jelly.

While some sources say to National Center for Food Preservation cook the apple jelly to 220°F (104°C), every time I've made this, it jelled at around 230°F (110°C). So it's best to use a thermometer but also to test your jelly by dropping a dab on a chilled plate, putting it in the freezer for a few minutes, then checking to see if the mixture has jelled by nudging it and seeing if it mounds and wrinkles, as

shown in the post. If you don't have a candy thermometer, you can use the "nudge" method to test your jelly.

One pound (450g) of apples cooked will yield about 1 cup (250ml) strained juice from the cooked apples. So if you have fewer apples, or you get a different yield (since all apples are different), you can use that as a guideline and add 3/4 cup (150g) sugar and 1 1/2 teaspoons of lemon juice per cup of strained apple juice. You can easily halve this recipe, too.

Note that in step #3 depending on the size of your strainer or colander, you may need to use two, as I do. One tip is when putting the apples in the strainers if you can't get them all in, after filling them up, let the mixture sit 5 to 10 minutes; it'll settle down and compress, and you should be able to add the rest after that.

Add the water, cover, and bring to a boil. When bubbling, reduce the heat a bit, leave the lid askew, and cook for 20 to 30 minutes, until the apples are tender and cooked through.

Note that in step #3 depending on the size of your strainer or colander, you may need to use two, as I do.

David

Line a mesh colander with a piece of muslin cloth or a few folds of cheesecloth (or use a jelly bag and stand) and set it over a deep bowl, then ladle the apples and the liquid into the colanders. (I use two lined colanders since it was quite a bit of apples.)

Let stand or at least 3 hours (but you'll get more juice if you let the apples drain around 8 hours), and during that time, no matter how tempting it looks,

do not press down

at any time on the apples to extract more juice or the jelly will get cloudy.

The next day, measure out the juice. You should have about 8 cups (2L) but may get a little more. Pour the juice into a large, non-reactive pot fitted with a candy thermometer, add the sugar and lemon juice, and bring to a boil. During cooking, as any white foam forms on the surface while the jelly is cooking, gently skim it off with a ladle. (See Note at the end of the recipe for some idea about repurposing the apples and the foam.)

Cook until the temperature reaches 220°F (104°C). At that point, turn off the heat and begin testing the jelly on a chilled plate in the freezer, using the method

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standard-article(title: [banana chocolate chip cake], author: [deb], source-name: [Smitten Kitchen], images: (), paragraphs: ([Somehow, despite how impossible it seems (to me, a person who has neither aged nor matured a day), it's been almost twenty years since I first told you about my family's favorite coffee cake . It's tall, plush, crisp with a flaky layer of cinnamon sugar on top, studded with a quilt of chocolate chips and is downright, well, adorable when cut into cubes because they're a little wobbly. When one tumbles, it shakes off a little pfft of cinnamon sugar, like a pup coming in from today's blizzard. It's perfect. It needs no changes or updates.], [Read more »],), insert-map: (:), word-count: 101, edited-for-length: false, debug-mode: false,)

standard-article(title: [Moving climate interest to action], author: [Mary Hoff], source-name: [Ensia], images: (), paragraphs: ([It likely is no surprise to hear that people don't always do the good they could — even if they want to.], [style="font-weight: 400;">When it comes to supporting political action around climate change, that reality is particularly stark. Analysis of a series of surveys of American adults spanning three years recently found a marked gap between the willingness of respondents to take political action with respect to global warming and the extent to which they actually did act.], [style="font-weight: 400;">In a series of six surveys from March 2021 to October 2023, researchers from George Mason University Center for Climate Change Communication and the Yale Program on Climate Change Communication asked thousands of respondents from a representative sample of U. S. adults whether they were willing to undertake four political actions related to global warming: sign a petition, donate funds, volunteer or contact government officials. They also asked which of these actions individuals actually took part in. Far fewer actually acted than said they would be likely to.], [style="font-weight: 400;">Looking more closely at those who expressed strong willingness to act, the study found that those for whom intention and action aligned were more likely than

standard-article(title: [Remanufacturing workshop management in blockchain and digital twin-based cloud remanufacturing service platform], author: [Yan Wang], source-name: [PLOS ONE Feed], images: (), paragraphs: ([by Qin Xiang, Xugang Zhang, Yan Wang], [The cloud remanufacturing service platform is an online platform that provides various services and resources for the remanufacturing industry. It promotes the digital transformation of the industry, improves resource utilization efficiency, and reduces environmental impact. However, there are prominent issues in sharing remanufacturing data securely, monitoring the remanufacturing process opaquely, and dealing with high uncertainties in remanufacturing operations within the remanufacturing intelligent workshop, which is an important component of the platform. This paper combines blockchain technology and digital twin technology to study the management mode of the remanufacturing workshop under the cloud remanufacturing service platform. Firstly, the overall framework of the cloud remanufacturing service platform is constructed, and a dual-chain structure with expandable subchains for storing transaction data and workshop data is designed. Secondly, by connecting the remanufacturing workshop with smart contracts and digital twin, real-time monitoring, remote collaboration, and data analysis optimization of the remanufacturing workshop are achieved. Digital twin is applied to the remanufacturing process, and a digital twin-based remanufacturing workshop architecture is established. The key technologies involved are analyzed. Finally, the management mode of the automotive remanufacturing workshop under the cloud-based remanufacturing service platform based on blockchain and digital twin is explored.],), insert-map: (:), word-count: 202, edited-for-length: false, debug-mode: false,)

standard-article(title: [Valorization of waste pharmaceutical residues via pyrolysis: Simultaneous production of biochar for Cd²⁺ removal and high-quality bio-oil/syngas], author: [Junchao Jia], source-name: [PLOS ONE Feed], images: (), paragraphs: ([by Zhizhen Feng, Bo Fu, Shanshan Han, Hong Yan, Puyang Feng, Wenxiao Li, Tao Qin, Tongtong Wang, Xinjie Zhang, Junchao Jia], [For the efficient utilization of pharmaceutical waste resources, tuber biochar (TB) and herbal biochar (HB) were prepared via oxygen-limited slow pyrolysis at 500 °C for 3 h, using residues from tuber-type Xinsuning capsule and herbal-type Changyanning pill as the raw materials, respectively. The biochars were characterized by FESEM, BET, XRD and FTIR, and the feedstock physico-chemical properties were measured by common agricultural chemical analysis methods. The results revealed that both biochars possessed a high percentage of elemental O, a honeycomb-like porous structure, and surfaces enriched with functional groups such as hydroxyl, carboxyl, and carbonyl. HB exhibited a larger specific surface area and pore volume than TB, making it a more recommended carbon material. The chemical compositions of the pyrolysis by-products were systematically analyzed. The bio-oils were rich in ketones, alkanes, alcohols, olefins, fatty acids, phenols, and heterocyclic

their non-acting counterparts and other Americans generally to talk and hear about global warming, acknowledge norms supporting climate action, and believe collective action makes a difference.], [The planet's most important stories — delivered to your inbox], [Be the first to hear about important new environmental stories. Sign up now to receive our newsletter.], [style="font-weight: 400;"]>What does this suggest regarding ways to close the "attitude-behavior gap"? The researchers noted that, although their study is not able to show a causal connection, it does hint at ways those seeking climate solutions might motivate the predisposed to align their actions with their predisposition. Suggestions include:], [style="font-weight: 400;"]>1) Ask people who are willing to take action to do so.], [style="font-weight: 400;"]>2) Make action easy.], [style="font-weight: 400;"]>3) Offer options so they can follow their personal preferences.], [style="font-weight: 400;"]>4) Point out how action supports their values and goals.], [style="font-weight: 400;"]>5) Provide examples of instances in which action has made a difference.], [style="font-weight: 400;"]>6) Talk about climate change.], [style="font-weight: 400;"]>7) Encourage others to talk about climate change.], [style="font-weight: 400;"]>8) Tell stories that normalize climate action.], [The post Moving climate interest to action appeared first on En-sia.],), insert-map: (:), word-count: 414, edited-for-length: false, debug-mode: false,)

standard-article(title: [My search for the perfect Sachertorte in Vienna], author: [Susan Smillie], source-name: [The Guardian Travel], images: (), paragraphs: ([The luscious chocolate and apricot torte is the stuff of legend in the grand, old world of Viennese coffeehouses. But which makes the tastiest?], [I'm on a tram on Vienna's Ringstrasse as towering facades, columns, statues and domes drift past, each more ornate than the last. Here, the State Opera; there, the Austrian parliament, built in the Greek neoclassical style.], [As I gawp, I shove cake in my mouth. After all, Vienna isn't just the city of music, or lavish architecture. Thanks, in part, to its centuries-old coffeehouse culture, it's also one of Europe's finest pastry destinations. Cake (or more precisely, torte, kuchen or Mehlspeisen) has its own day here – "Sweet Friday", the most delicious of Catholic customs, when meat dishes are replaced with sweets. I have been introduced to it via the medium of Marillenknödel – apricot dumplings.], [Continue reading...],), insert-map: (:), word-count: 143, edited-for-length: false, debug-mode: false,)

compounds, identifying them as potential sources of liquid fuels and chemical feedstocks. The most abundant components in bio-oils from tuber and herbal biomass were "Ethanol, 2,2-diethoxy-" (7.25%) and "Phosphonic acid, (p-hydroxyphenyl)-" (10.52%), respectively. The syngas has a low hydrogen content, is mainly pyrolysis off-gas and therefore has a limited application potential. Furthermore, the environmental application for Cd²⁺ removal was critically evaluated. Adsorption isotherms demonstrated high adsorption capacities, well-described by the Freundlich model ($R^2 \geq 0.99$), indicating multilayer adsorption. The maximum adsorption capacities for TB and HB were 188.89 and 186.67 mg·g⁻¹, respectively. Kinetic studies revealed that the adsorption process followed the Elovich model ($R^2 \geq 0.98$), suggesting heterogeneous diffusion, with HB achieving a higher equilibrium capacity (85.67 mg·g⁻¹) than TB (73.70 mg·g⁻¹). In conclusion, pyrolysis, particularly using herbal biomass, presents a promising strategy for the comprehensive and high-value utilization of waste pharmaceutical residues, simultaneously producing effective adsorbents for heavy metal remediation and valuable bio-energy products.],), insert-map: (:), word-count: 317, edited-for-length: false, debug-mode: false,)

standard-article(title: [Emission-reduction cooperation among technologically complementary manufacturers under a carbon cap-and-trade mechanism with price competition], author: [Xiangwei Gui], source-name: [PLOS ONE Feed], images: (), paragraphs: ([by Mingke He, Yi Zhou, Yihang Qiao, Yingzhe Ding, Xiangwei Gui], [In recent years, driven by emission reduction targets, an increasing number of manufacturers producing similar products have been compelled to seek emission reduction cooperation even while competing in the market, a phenomenon that has attracted growing attention in recent studies. Based on the carbon cap-and-trade mechanism, this study develops a noncooperative-cooperative biform game model to examine the optimal decisions of technologically complementary manufacturers engaging in emission reduction cooperation under price competition. The model describes alliance profits using a characteristic function and applies the Shapley value for profit allocation, while equilibrium outcomes are derived through noncooperative game analysis. The research results show that the high carbon emission reduction investment coefficient will inhibit the carbon abatement development level of manufacturers and reduce their profits. Without a carbon cap-and-trade mechanism, competitive manufacturers lack incentives for technological collaboration. An increase in carbon trading prices significantly promotes emission reductions and profit growth, whereas the effect of government-allocated initial carbon allowances remains limited. Moreover, the improvement of manufacturers' technological conversion capability can improve the level of carbon abatement development and enhance their profitability under price competition. In general, this study provides theoretical and practical guidance for the cooperative emission reduction of competitive enterprises under the carbon emission

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Painting with odors: How olfactory stimuli influence artistic expression, emotional response, visual perception, and object selection

Nobuyuki Sakai · PLOS ONE Feed

by Zahra Davoudi, Nobuyuki Sakai

In recent years, the relationship between olfaction and vision has received increasing attention. We combined psychology and art to investigate how two specific odors, strawberry and rose, influence visual perception, artistic expression, creativity, and object selection across two studies. In Study 1, 24 participants created paintings inspired by rose and strawberry odors. Sixty additional participants evaluated these paintings using seven semantic differential scales. In the final phase, another 60 participants rated the odors using the same scales. In Study 2, 60 participants were randomly assigned to one of three rooms: a strawberry-odorized room, a rose-odorized room, or a control room with no odor. Five artificial objects (strawberries, lemons, and roses) were placed on a table in each room. Without being told about the presence of odors, the participants were asked to select one object and paint it. Afterward, they reported the odor they perceived and selected the associated colors using a standardized color panel. This design aimed to determine whether specific odors influenced object selection and visual perception. Together, these experiments provide converging evidence suggesting consistent odor-color associations and indicate that odors may influence visual perception and object selection.

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ANALYSIS

Nelson Elhage, Nelson Elhage: I have spent many years as a software engineer who was a total outsider to machine-learning, but with some curiosity and occasional peripheral interactions with it. During this time, a recurring theme for me was horror (and, to be honest, disdain) every time I encountered the widespread usage of Python pickle in the Python ML ecosystem. In addition to their major security issues¹, the use of pickle for serialization tends to be very brittle, leading to all kinds of nightmares as you evolve your code and upgrade libraries and Python versions.

Nelson Elhage, Nelson Elhage: I'm at Dragon*Con this weekend, my second time here now. I decided that if I was going to Dragon*Con again, I needed to do something in terms of costuming, and I wanted it to be something unique – I wasn't going to come anywhere near as epic as some of the costumes people pull off, but I wanted something that was going to be a little impressive, hopefully totally unique, and perhaps slightly insane.

deb, Smitten Kitchen: `class="smittenkitchen-print-hide">`

Unless you're living your life better than me (probably!), I bet it's been way too long since you last had a baked potato for dinner — or, as they're more charmingly called across the pond, "jacket potato." And it's a crime because they're so cozy and uncomplicated to make, we could fix this right now.

[Read more »](#)

Nelson Elhage, Nelson Elhage: I love VMware workstation. I keep VMs around for basically every version of every major Linux distribution, and use them heavily for all kinds of kernel testing and development. This post is a quick writeup of my networking setup with VMware Workstation, using dnsmasq to assign my VMs addresses and provide a DNS server to resolve VM addresses. The objective I want to be able to resolve my VM's hostnames so that I can ssh to them, or run other network services and access them from the host.

Nelson Elhage, Nelson Elhage: If you're familiar with nearly any mainstream programming language, and I asked you to draw a diagram of an array, the array indices, and the array elements, odds are good you'd produce a diagram something like this: In this post, I want to persuade you to replace that image, or, at least, to augment it with an alternate view on the world. I want to argue that, rather than numbering elements of an array, it makes just as much sense, and in many cases more, to number the spaces between elements:

Nelson Elhage, Nelson Elhage: Last week, I wrote about the mindset that computer systems can be understood, and behaviors can be explained, if we're willing to dig deep enough into the stack of abstractions our software is built atop. Some of the ensuing discussion on Twitter and elsewhere lead me to write this followup, in which I want to run through a few classes of systems where I've found pursuing in-detail understanding of the system wasn't the right answer.

Nelson Elhage, Nelson Elhage: Software engineers, as a rule, suck at writing things down. Part of this is training – unlike chemists and biologists who are trained to obsessively document everything they do in their lab notebooks, computer scientists are taught to document the end results of their work, but aren't, in general, taught to take notes as they go, and document the steps they take in building a system. 6.005, MIT's new introductory software engineering class, attempted to require its students to keep lab notebooks for a few semesters, and was met with near-universal complaints and ridicule from the students ("Lab notebooks?

Nelson Elhage, Nelson Elhage: (This post is cross-posted from Honeycomb's instrumentation series). One of my favorite concepts when thinking about instrumenting a system to understand its overall performance and capacity is what I call "time utilization". By this I mean: If you look at the behavior of a thread over some window of time, what fraction of its time is spent in each "kind" of work that it does? Let's make this notion concrete by examining a toy example.

Nelson Elhage, Nelson Elhage: At this point in my career, I've worked on at least three projects where performance was a defining characteristic: Livegrep, Taktician, and Sorbet (I discussed sorbet in particular last time, and livegrep in an earlier post). I've also done a lot of other performance work on the tools I use, some of which ended up on my other blog, Accidentally Quadratic. In this post, I want to reflect on some of the lessons I've learned while writing performant software, and working with rather a lot more not-so-performant software.

Nelson Elhage, Nelson Elhage: I haven't been feeling very motivated to blog lately – I've missed the last two weeks of Iron Blogger, and I'm not totally enthusiastic about any of the items on my "to blog" list. But, I do enjoy blogging when I actually get into posts, and I'd like to keep updating this blog. So, in a bit of a copout, and following in Edward's footsteps, this is an appeal to all of you: What should I blog about?

Nelson Elhage, Nelson Elhage: While talking about thinking about tests and testing in software engineering recently, I've come to the conclusion that there are (at least) two major ideas and goals that people have when they test or talk about testing. This post aims to outline what I see as these two schools, and explore some reasons engineers coming from these different perspectives can risk talking past each other. Two reasons to test Testing for correctness The first school of testing comprises those who see testing as a tool for validating a software artifact against some externally-defined standard of correctness.

BBC Science & Environment, BBC Science & Environment: Nasa is on track to launch its Artemis II mission in April which will be tracked by Goonhilly Earth Station, near Helston.

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